DATA SCIENCE INTERNSHIP PROGRAM

SALES DATA ANALYSIS & VISUALIZATION

AN INTERNSHIP PROJECT REPORT

Submitted by

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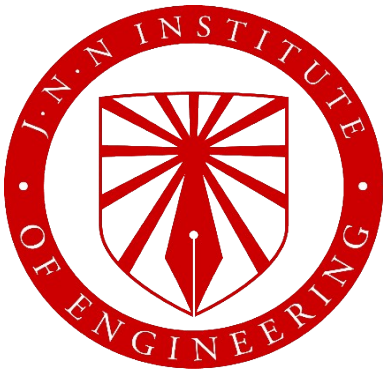
in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science



J.N.N Institute of Engineering

(An Autonomous Institute , Affiliated to Anna University, Chennai)

AUGUST – 2026

VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

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IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.

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- **DM 3:** To make the students industry ready and to enhance their employability through training, internships and real-time
- **DM 4:** To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry

PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
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PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team

m

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight , hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.



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BONAFIDE CERTIFICATE

Certified that this internship project report
"SALES DATA ANALYSIS AND VISUALIZATION"
(Data Science Internship) is the bonafide work of "SHARON BLESSY R V"
(Reg No: 110725102229) who carried out the project under my supervision.

Signature of the HOD with date

Dr. P. Nagarajan

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Signature of the Supervisor with date

Ms. M Gayathri Devi

Project Supervisor,
Department of Artificial Intelligence
& Data Science Engineering,
J.N.N Institute of Engineering, Chennai

Certified that the above students have attend of viva voce during the exam held on…….

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

First and foremost, we express our sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.,** and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYAHANDRAN M.S.,** for providing necessary facilities for the completion of our project.

We thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.,** for the encouragement provided with the congenial environment essential for us to complete the project.

We take this opportunity to express our gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving us continuous encouragement for completing our project.

We extend our sincere thanks to **VEI Technologies Pvt. Ltd., Chennai** and its **Founder & CEO, Dr. B. Ezhilavan,** for giving us the opportunity to undergo this Data Science internship, and we sincerely thank our **project supervisor** and the entire Data Science team at VEI Technologies for their guidance, patience and technical support throughout this internship.

Finally, we would like to thank our parents and friends for their constant support and encouragement.

ABSTRACT

Data is now central to every decision taken in business, and data science is the discipline that turns raw data into useful knowledge. This report documents the work carried out during the Data Science Internship at VEI Technologies Pvt. Ltd., Chennai, completed between 3rd August 2026 and 26th August 2026.

The internship project, titled “Sales Data Analysis and Visualization”, was built in seven connected modules. Module 1 prepared and inspected the raw sales data. Module 2 cleaned the data with Python and pandas by removing duplicate records, standardising labels, handling missing values and converting dates. Module 3 calculated the main sales indicators. Module 4 studied the monthly and quarterly sales trend. Module 5 analysed sales by category, product and region. Module 6 studied customer segments and the effect of discounts on profit. Module 7 presented the findings through a heat map, a dashboard and an Excel report. A set of eleven automatic checks confirmed that the totals agree in every view of the data.

The analysis of a sample data set of 1,194 cleaned orders showed total sales of Rs. 5,589,069.15 and a profit margin of 23.72 %. Sales were highest in the last quarter, and the profit margin fell from 29.90 % for orders without discount to -2.88 % for orders with a 21–30 % discount. Together, the project demonstrates the complete data science workflow – collection, cleaning, analysis, visualization and reporting – on a business problem.

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CHAPTER 1
INTRODUCTION

1.1 Overview

Every business, whether it is a small shop or a large retail chain, records each sale it makes: what was sold, to whom, in which region, on which day and at what price. When these records are collected over a year they become a very valuable source of knowledge. They can show which products earn the most profit, which months are busy, which regions need more attention and whether the discounts given to customers are really helping the business. However, a sales file with thousands of rows cannot be understood by simply looking at it. It must be cleaned, summarised and presented in a form that a manager can read in a few minutes.

This report presents the project “Sales Data Analysis and Visualization”, which was developed during the Data Science Internship at VEI Technologies Pvt. Ltd., Chennai, from 3rd August 2026 to 26th August 2026. The project takes a raw retail sales data set, cleans it with Python and pandas, calculates the main sales indicators, studies the sales by month, category, product, region, customer segment and discount level, and finally presents the findings through charts, a dashboard and an Excel report.

The project therefore covers the main steps of a data science workflow – data collection, data cleaning, exploratory analysis, visualization and reporting – on one realistic business problem. It also gave the intern practical exposure to the tools that are widely used in the industry for this work.

1.2 Internship Details

Table 1.1 – Details of the internship

Particulars	Details
Name of the Intern	SHARON BLESSY R V
Register Number	110725102229
Department	Artificial Intelligence & Data Science
Institution	J.N.N Institute of Engineering, Chennai
Organization	VEI Technologies Pvt. Ltd., Chennai
Field of Internship	Data Science
Role	Data Science Intern
Duration	24 days (03.08.2026 to 26.08.2026)
Project Title	Sales Data Analysis and Visualization
Certificate No.	VEI0820261532
Certificate Issued By	Dr. B. Ezhilavan, Founder & CEO

VEI Technologies Pvt. Ltd. is an AICTE, MSME and ISO 9001:2015 certified institution located in Chennai. During the internship the intern worked on the project described in this report and was awarded a Certificate of Completion, a copy of which is given in the Appendix.

1.3 Objectives

- To understand the data science workflow and the role of Python in business data analysis.
- To load a raw sales data set into pandas and to inspect its structure, data types and quality.
- To clean the data by removing duplicate records, standardising text labels, handling missing values and invalid entries.
- To calculate key sales indicators such as total sales, total profit, profit margin, average order value and units sold.
- To analyse sales by month, category, product, region, customer segment and discount level, and to find the strong and weak areas of the business.
- To present the results with clear charts, a dashboard and an Excel report that can be shared with non-technical readers.

1.4 Overview of the Project Modules

The project was divided into seven connected modules. Each module takes the output of the previous one, so the work moves step by step from the raw file to the final report. A short eighth script tests the results (Section 4.8).

- **Module 1 – Setup and Data Collection:** project folder, preparation of the sample sales data set and first inspection with pandas.
- **Module 2 – Data Cleaning:** removal of duplicates, standardisation of labels, treatment of missing values, date conversion and removal of invalid rows.
- **Module 3 – Sales KPIs:** total sales, profit, margin, orders, units, average order value and descriptive statistics.
- **Module 4 – Sales Trend Analysis:** monthly and quarterly sales, growth percentage, best and weakest month.
- **Module 5 – Category, Product and Region Analysis:** sales and profit by category, top products and regional share.
- **Module 6 – Customer, Segment and Discount Analysis:** customer segments, effect of discount on profit margin, top customers.
- **Module 7 – Dashboard and Excel Report:** heat map, four-panel dashboard and a multi-sheet Excel report.



Fig. 1.1 – Phases of the sales analysis project

1.5 Problem Statement

In many small and medium businesses the sales records are kept in large spreadsheets that are filled in by several people. Such files usually contain repeated entries, spelling and case differences in

names, missing values and wrong numbers. When the manager wants to know which product or region is doing well, the figures must be added by hand, which is slow and easily leads to mistakes.

A second difficulty is that totals alone do not explain the business. A high sales figure does not always mean a high profit, and a heavy discount may increase the number of orders while reducing the profit. A method is therefore needed that can clean the sales data, calculate the indicators correctly and show the results visually so that decisions can be taken on facts.

1.6 Scope of the Project

The project was carried out on a sample retail sales data set for the calendar year 2025. The scope of the work is as follows.

- **Included:** cleaning and validation of the raw sales data using Python and pandas.
- **Included:** calculation of sales indicators and analysis by time, category, product, region, segment and discount.
- **Included:** static charts, a heat map, a dashboard image and an Excel report.
- **Included:** a set of automatic checks that confirm that the totals in every view of the data agree.
- **Not included:** a web or mobile application, live connection to a company database, sales forecasting with machine learning models.

1.7 Organisation of the Report

The report is organised into six chapters. Each chapter builds on the previous one.

- **Chapter 1 – Introduction:** states the purpose of the internship, the objectives, the modules and the scope.
- **Chapter 2 – Literature Review:** explains the concepts and tools used and the limits of the existing approaches.
- **Chapter 3 – Methodology:** describes the approach, the data set, the weekly schedule, the workflow and the requirements.
- **Chapter 4 – Implementation:** presents the seven modules with code, output and figures, followed by testing and challenges.
- **Chapter 5 – Results and Discussion:** discusses the numerical results, the skills gained and the overall experience.
- **Chapter 6 – Conclusion and Future Work:** summarises the work, its limits and the possible improvements.

CHAPTER 2
LITERATURE REVIEW

2.1 Data Science and Sales Analytics

Data science is the field that uses programming, statistics and domain knowledge to turn data into useful decisions. A typical project moves through the stages of collecting data, preparing it, exploring it, building models or summaries, and communicating the results [1]. Sales analytics is one of the most common applications of data science in business. It studies past sales to explain what has happened and to guide what should be done next.

Sales analytics is usually descriptive in the beginning: it answers questions such as “how much did we sell?”, “which product earns the most?” and “when are the busy months?”. Only after these questions are answered clearly can more advanced work such as forecasting be attempted. This internship project concentrates on the descriptive stage, which forms the foundation of every later analysis.

2.2 Data Cleaning and Preparation

Real data is rarely perfect. Duplicate rows, missing values, inconsistent spellings and impossible numbers are common, and any analysis done on such data can give wrong answers. Data cleaning is the process of finding and correcting these problems before analysis begins. It is widely accepted that data preparation takes a large part of the time in a data science project [1].

In Python the pandas library offers ready-made tools for this stage. The functions drop_duplicates, fillna, str.strip, str.title and to_datetime are enough to solve most of the problems found in a sales file [2]. The important point is that every cleaning step should be recorded in code so that it can be repeated on a new file in the same way.

2.3 Exploratory Data Analysis

Exploratory data analysis (EDA) is the practice of looking at a data set from many sides, using summary numbers and simple charts, before any formal conclusion is drawn [3]. In EDA the analyst groups the data, calculates totals and averages, compares categories and looks for patterns and unusual values. The group-by and pivot-table operations of pandas are the main tools for this work, and they correspond closely to the pivot tables that many people already know from spreadsheets.

2.4 Data Visualization and Dashboards

A chart can show a pattern that is hard to see in a table of numbers. A line chart is best for a trend over time, a bar chart for comparing categories, a pie chart for the share of a whole and a heat map for comparing two categories at the same time. Good visualization follows simple rules: each chart has a clear title, labelled axes and a suitable chart type for the question being answered [3].

A dashboard places several of these charts on one page so that the reader can understand the position of the business at a glance. In Python the Matplotlib and Seaborn libraries are used to draw

such charts [4], [5].

2.5 Key Sales Performance Indicators

Table 2.1 lists the indicators that were calculated in this project, with their formulas.

Table 2.1 – Sales indicators used in the project

Indicator	Formula	Meaning
Total sales	Sum of sales of all orders	Total money earned from customers
Total profit	Sum of (sales – cost)	Money left after the cost of goods
Profit margin (%)	Total profit ÷ Total sales × 100	Profit earned on every 100 rupees of sales
Average order value	Total sales ÷ Number of orders	Typical value of one order
Units sold	Sum of quantity	Number of items sold
Growth (%)	(This month – Last month) ÷ Last month × 100	Change in sales from the previous month
Share (%)	Sales of a region ÷ Total sales × 100	Contribution of a region to total sales

2.6 Tools and Their Purpose

Table 2.2 – Tools and technologies used

Tool / Library	Purpose in the project
Python 3	Main programming language used for all the modules
pandas	Loading the CSV file, cleaning, grouping, pivot tables and statistics [2]
NumPy	Numerical operations and random number generation for the sample data [6]
Matplotlib	Line, bar, pie and dashboard charts [4]
Seaborn	Heat map of sales by region and category [5]
openpyxl	Writing the multi-sheet Excel report
Microsoft Excel	Opening and presenting the final report [7]
VS Code / Jupyter Notebook	Writing and running the Python scripts

2.7 Existing Approaches and Their Limitations

Before data science tools became common, sales analysis was done in two main ways, and both are still used in many small businesses.

- **Manual calculation:** the sales registers are added with a calculator. This is very slow, and one wrong entry changes every later total.
- **Spreadsheet only:** the data is placed in a spreadsheet and summarised with formulas and pivot tables. This is better, but the cleaning steps are done by hand, they are not recorded and

they must be repeated each time a new file arrives.

- **Fixed business-intelligence reports:** commercial tools give ready dashboards but they are costly, need set-up and are difficult to change for a new question.

2.8 Need for the Approach Followed

A Python-based approach solves these limits. The whole process – cleaning, calculation and charting – is written as code, so it can be run again in seconds on a new data file and it gives the same result every time. Each step is visible, so a reader can check how a number was obtained. Python is free, it has a very large community and it connects easily with Excel for the final report. For these reasons the Python and pandas approach was chosen for the internship project.

CHAPTER 3
METHODOLOGY

3.1 Overall Approach

The project follows a layered approach, shown in Fig. 3.1. The raw sales file enters at the data source layer. The processing layer cleans it and prepares new columns. The analysis layer calculates indicators and groups the data in different ways. Finally the presentation layer turns the results into charts, a dashboard and an Excel report.

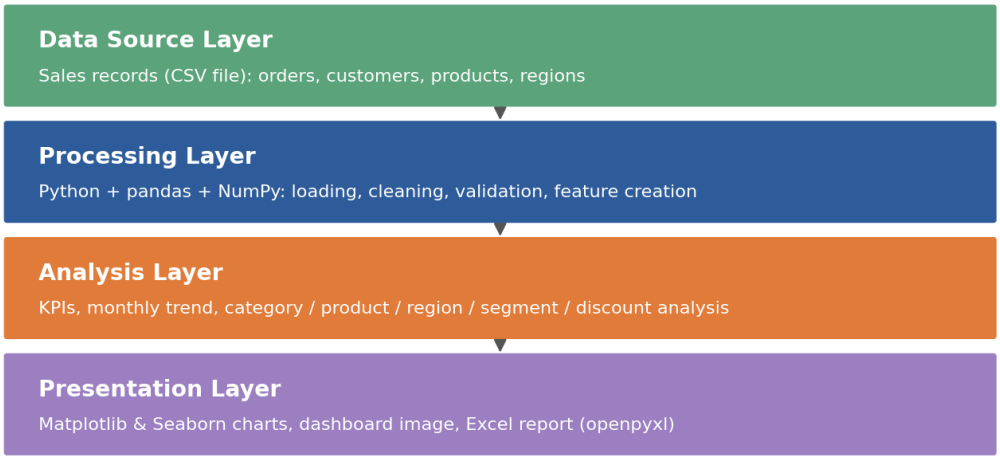


Fig. 3.1 – Layered architecture of the sales analysis project

3.2 Data Set Description

The real sales records of a company are confidential and cannot be used in a student report. The analysis was therefore carried out on a sample retail sales data set for the year 2025. The data set was prepared with a Python script (see Code 4.1) that uses a fixed random seed, so the same file can be produced again. The raw file contains **1,218 records** and 12 columns, and it was deliberately given the kinds of errors that real files contain – duplicate rows, mixed upper and lower case labels, extra spaces, missing values and zero quantities. The columns are described in Table 3.1.

Table 3.1 – Columns of the sales data set

Column	Type	Description
order_id	Text	Unique number of the order
order_date	Date	Date on which the order was placed (dd-mm-yyyy)
customer_id	Text	Code of the customer
segment	Text	Customer segment: Consumer, Corporate or Home Office
region	Text	Sales region: North, South, East or West
category	Text	Product category: Electronics, Furniture, Stationery or Clothing
product	Text	Name of the product sold
unit_price	Number	List price of one item in rupees

Column	Type	Description
quantity	Number	Number of items in the order
discount	Number	Discount given, as a fraction (0.10 = 10 %)
sales	Number	Amount charged to the customer after discount
profit	Number	Sales minus the cost of the goods

3.3 Week-wise Internship Schedule

The 24 days of the internship were planned in four stages, as shown in Table 3.2.

Table 3.2 – Week-wise schedule of the internship

Week / Dates	Work area	Activities
Week 1 (03.08 – 08.08)	Orientation and setup	Introduction to data science, Python and pandas basics, understanding the sales data set, project folder (Module 1)
Week 2 (10.08 – 15.08)	Cleaning and KPIs	Quality check, cleaning of the raw data (Module 2), calculation of sales indicators (Module 3)
Week 3 (17.08 – 22.08)	Analysis	Trend analysis (Module 4), category, product and region analysis (Module 5), customer and discount analysis (Module 6)
Week 4 (24.08 – 26.08)	Reporting	Dashboard and Excel report (Module 7), testing, preparation of the report and submission

3.4 Data Cleaning Rules

Six rules were applied in order, as shown in Fig. 3.2. Each rule corresponds to one problem found in the quality check of Module 2.

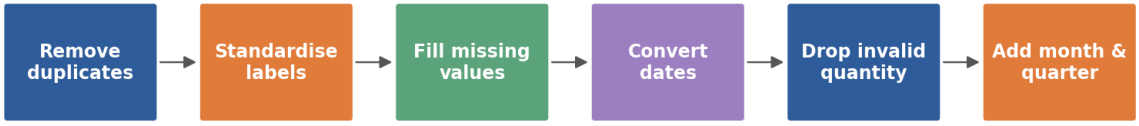


Fig. 3.2 – Data cleaning steps applied to the raw sales data

- **Duplicates:** rows that are exactly the same as an earlier row are removed.
- **Labels:** spaces are stripped and the text is converted to title case, so “ south ”, “south” and “South” become one label.
- **Missing values:** a missing discount is taken as 0 (no discount) and a missing segment is marked “Unknown”.
- **Dates:** the text dates in dd-mm-yyyy form are converted to real date values.
- **Invalid rows:** rows with quantity 0 are not real orders and are removed.
- **New columns:** month, month name and quarter are added to make time analysis easy.

3.5 Analysis Methods

The cleaned data was studied with the following methods.

- **Descriptive statistics:** count, mean, standard deviation, minimum, quartiles and maximum of the numeric columns.
- **Group-by aggregation:** totals of sales and profit for every month, quarter, category, product, region and segment.
- **Growth calculation:** month-over-month percentage change in sales.
- **Binning:** the discount was divided into four bands (0 %, 1–10 %, 11–20 % and 21–30 %) to compare profit margins.
- **Pivot table and heat map:** sales for every region and category pair.

3.6 System Requirements

No special equipment is needed. Table 3.3 lists the minimum requirements for running the project.

Table 3.3 – Minimum hardware and software requirements

Item	Requirement
Processor	Intel Core i3 or equivalent, or higher
Memory	4 GB RAM (8 GB recommended)
Storage	500 MB of free disk space
Operating system	Windows 10 / 11, Linux or macOS
Language	Python 3.10 or later
Libraries	pandas, NumPy, Matplotlib, Seaborn, openpyxl
Editor	VS Code, Jupyter Notebook or any text editor
Other software	Microsoft Excel or a compatible spreadsheet program

3.7 Functional and Non-Functional Requirements

Functional requirements

- The system shall read the raw sales file and report its size, columns and data types.
- The system shall clean the data and save a clean copy of the file.
- The system shall calculate the sales indicators and the grouped summaries.
- The system shall draw the charts and save them as image files.
- The system shall write the summaries into an Excel workbook with separate sheets.

Non-functional requirements

- **Accuracy:** the totals shown in every view of the data must agree with each other.
- **Repeatability:** running the scripts again on the same file must give the same results.
- **Readability:** the code is divided into small modules with clear names and the charts have titles and labels.
- **Simplicity:** the project uses only free and widely available tools.

3.8 Data Processing Workflow

Fig. 3.3 shows how the data moves through the project. The raw CSV file is loaded with pandas, cleaned and validated and saved as a clean file. All the later modules read the clean file, so the cleaning is done only once.



Fig. 3.3 – Data processing workflow of the project

3.9 Testing Method

Because no single answer sheet exists for a sales analysis, the results were tested by consistency. A set of eleven automatic checks was written in Python (Module 8). They test the cleaning (no duplicates, standard labels, no missing values), the data types and, most importantly, that the total of the monthly, category and region views are all equal to the overall total. The Excel report is also checked against the analysed total. The checks and their results are given in Section 4.8.

CHAPTER 4
IMPLEMENTATION

This chapter explains the seven modules of the project. For each module the services (tools) used are listed first, then the working is explained, the Python code is shown and the output obtained on running the code is given as a figure. All the code was run on the sample data set described in Section 3.2.

4.1 Module 1: Project Setup and Data Collection

Services used: Python 3, pandas, NumPy, VS Code.

The first task was to create a project folder with separate places for the data, the figures and the output (Fig. 4.1). The sample data set was then generated by the script in Code 4.1. The script chooses the month, product, quantity, discount, region and segment of each order with a random generator that has a fixed seed (2026), so the same file is produced every time. It then adds the data-quality problems that real files have.

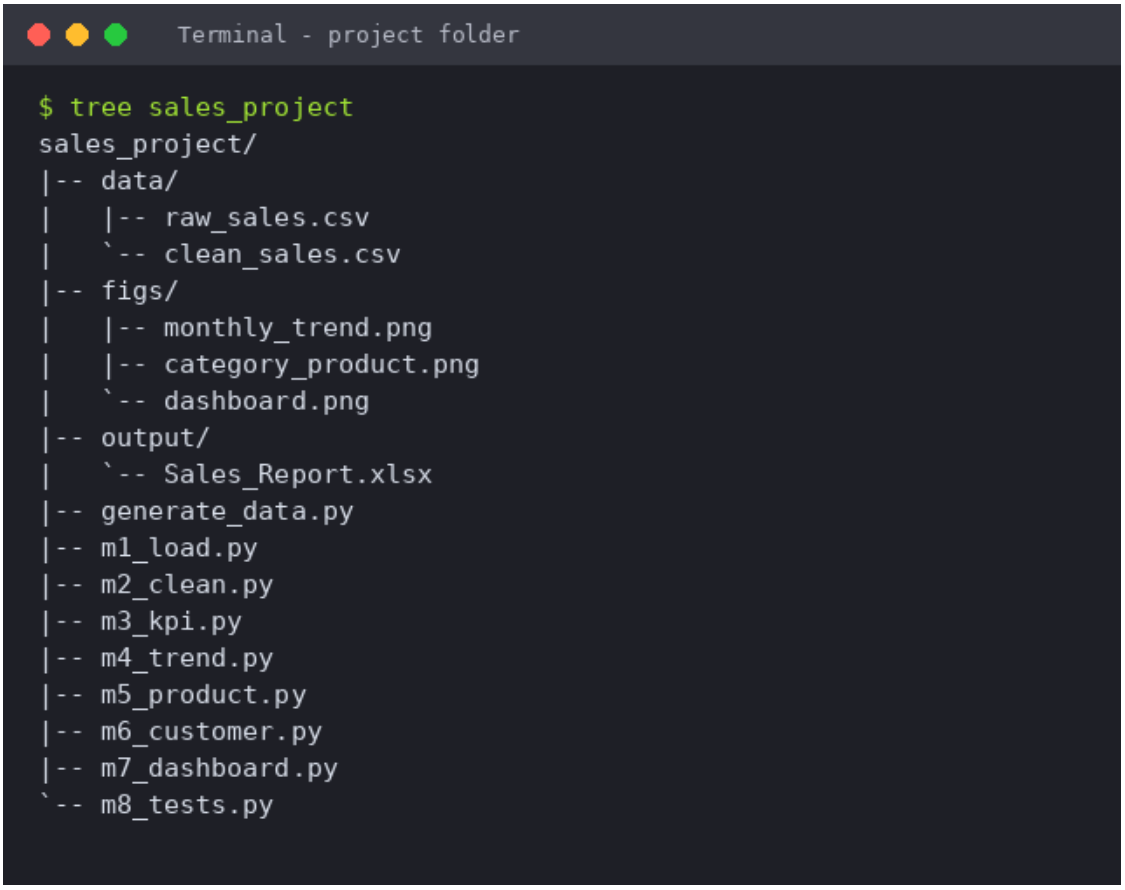


Fig. 4.1 – Project folder structure

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(2026)
N = 1200
cats = {
    "Electronics": [("Wireless Mouse", 899, 0.28),
```

```
                     ("Bluetooth Speaker", 2499, 0.30),
                     ("Headphones", 1999, 0.32), ("Power Bank", 1299, 0.27),
                     ("Smart Watch", 3999, 0.25)],
    "Furniture": [("Office Chair", 5499, 0.22), ("Study Table", 4299, 0.24),
                  ("Bookshelf", 3299, 0.26), ("Desk Lamp", 799, 0.35)],
    "Stationery": [("Notebook Pack", 249, 0.40), ("Gel Pen Box", 149, 0.45),
                   ("File Folder", 199, 0.42), ("Marker Set", 329, 0.38)],
    "Clothing": [("Cotton T-Shirt", 599, 0.45), ("Denim Jeans", 1799, 0.38),
                 ("Hooded Jacket", 2299, 0.36), ("Sports Shoes", 2799, 0.33)],
}
regions = {"North": 0.18, "South": 0.34, "East": 0.20, "West": 0.28}
segments = {"Consumer": 0.52, "Corporate": 0.30, "Home Office": 0.18}
month_w = np.array([0.7, 0.72, 0.8, 0.85, 0.9, 0.88,
                    0.82, 0.9, 1.0, 1.25, 1.45, 1.6])
month_w /= month_w.sum()

rows = []
for i in range(N):
    m = rng.choice(12, p=month_w) + 1
    d = rng.integers(1, 28)
    date = pd.Timestamp(2025, m, d)
    cat = rng.choice(list(cats), p=[0.30, 0.18, 0.27, 0.25])
    prod, price, margin = cats[cat][rng.integers(len(cats[cat]))]
    qty = int(rng.integers(1, 6))
    disc = float(rng.choice([0, 0.05, 0.10, 0.15, 0.20, 0.30],
                            p=[0.35, 0.2, 0.2, 0.12, 0.09, 0.04]))
    region = rng.choice(list(regions), p=list(regions.values()))
    seg = rng.choice(list(segments), p=list(segments.values()))
    cust = f"C{rng.integers(1, 260):04d}"
    sales = round(price * qty * (1 - disc), 2)
    profit = round(sales - price * (1 - margin) * qty, 2)
    rows.append([f"ORD{1001 + i}", date.strftime("%d-%m-%Y"), cust, seg, region,
                 cat, prod, price, qty, disc, sales, profit])

cols = ["order_id", "order_date", "customer_id", "segment", "region", "category",
        "product", "unit_price", "quantity", "discount", "sales", "profit"]
df = pd.DataFrame(rows, columns=cols)

# inject realistic data-quality problems
idx = rng.choice(N, 30, replace=False)
df.loc[idx[:12], "region"] = df.loc[idx[:12], "region"].str.lower()
df.loc[idx[12:20], "region"] = " " + df.loc[idx[12:20], "region"] + " "
df.loc[idx[20:30], "category"] = df.loc[idx[20:30], "category"].str.upper()
miss = rng.choice(N, 25, replace=False)
df.loc[miss[:15], "discount"] = np.nan
df.loc[miss[15:], "segment"] = np.nan
df.loc[rng.choice(N, 6, replace=False), "quantity"] = 0
dups = df.sample(18, random_state=5)
df = pd.concat([df, dups]).sample(frac=1, random_state=9).reset_index(drop=True)
df.to_csv("data/raw_sales.csv", index=False)
print(df.shape)
```

Code 4.1 – Script that prepares the sample sales data (generate_data.py)

After the file was ready it was loaded with pandas and examined (Code 4.2). The program prints the shape of the data, the first five rows and the information table that shows the data type and the number of non-empty values of each column.

```
import pandas as pd

df = pd.read_csv("data/raw_sales.csv")
print("Shape of the raw data:", df.shape)
print()
cols = ["order_id", "order_date", "segment", "region",
        "category", "quantity", "sales"]
print(df[cols].head(5).to_string(index=False))
print()
df.info()
```

Code 4.2 – Loading and inspecting the raw data (m1_load.py)

```
Terminal - m1_load.py

$ python m1_load.py
Shape of the raw data: (1218, 12)

order_id order_date  segment region   category  quantity   sales
 ORD1604 12-08-2025 Consumer  South  Furniture         3 2397.00
 ORD1454 15-05-2025 Consumer   East  Furniture         1 2639.20
 ORD1838 26-10-2025 Consumer  South Stationery         4 1184.40
 ORD1552 20-08-2025 Consumer   East   Clothing         3 7977.15
 ORD1990 18-10-2025 Consumer  South Stationery         2  448.20

<class 'pandas.DataFrame'>
RangeIndex: 1218 entries, 0 to 1217
Data columns (total 12 columns):
...
```

Fig. 4.2 – Loading and previewing the raw sales data

```
Terminal - m1_load.py

$ python m1_load.py   (continued)
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   order_id     1218 non-null   str
 1   order_date   1218 non-null   str
 2   customer_id  1218 non-null   str
 3   segment      1208 non-null   str
 4   region       1218 non-null   str
 5   category     1218 non-null   str
 6   product      1218 non-null   str
 7   unit_price   1218 non-null   int64
 8   quantity     1218 non-null   int64
 9   discount     1202 non-null   float64
 10  sales        1218 non-null   float64
 11  profit       1218 non-null   float64
dtypes: float64(3), int64(2), str(7)
memory usage: 114.3 KB
```

Fig. 4.3 – Structure of the raw data set (columns, types and non-null counts)

The information table already points to problems: the segment column has 1,208 non-null values and the discount column has 1,202 out of 1,218 rows, so some values are missing. These problems are treated in the next module.

4.2 Module 2: Data Cleaning

Services used: Python 3, pandas.

The cleaning module first prints a quality check of the raw data and then applies the six rules of Section 3.4. Code 4.3 shows the complete code. A quality check is printed again after cleaning to prove that every problem has been solved.

```
import pandas as pd

raw = pd.read_csv("data/raw_sales.csv")
print("---- Quality check BEFORE cleaning ----")
print("Rows                :", len(raw))
print("Duplicate rows      :", raw.duplicated().sum())
print("Missing values      :")
print(raw.isnull().sum()[raw.isnull().sum() > 0].to_string())
print("Region label variants   :", raw["region"].nunique())
print("Category label variants :", raw["category"].nunique())
print("Zero quantity rows  :", (raw["quantity"] == 0).sum())

df = raw.drop_duplicates().copy()
df["region"] = df["region"].str.strip().str.title()
df["category"] = df["category"].str.strip().str.title()
df["segment"] = df["segment"].fillna("Unknown")
df["discount"] = df["discount"].fillna(0)
df["order_date"] = pd.to_datetime(df["order_date"], format="%d-%m-%Y")
df = df[df["quantity"] > 0]
df["month"] = df["order_date"].dt.month
df["month_name"] = df["order_date"].dt.strftime("%b")
df["quarter"] = "Q" + df["order_date"].dt.quarter.astype(str)
df.to_csv("data/clean_sales.csv", index=False)

print()
print("---- Quality check AFTER cleaning ----")
print("Rows                :", len(df))
print("Duplicate rows      :", df.duplicated().sum())
print("Missing values      :", int(df.isnull().sum().sum()))
print("Region label variants   :", df["region"].nunique())
print("Category label variants :", df["category"].nunique())
print("Zero quantity rows  :", (df["quantity"] == 0).sum())
```

Code 4.3 – Quality check and data cleaning (m2_clean.py)

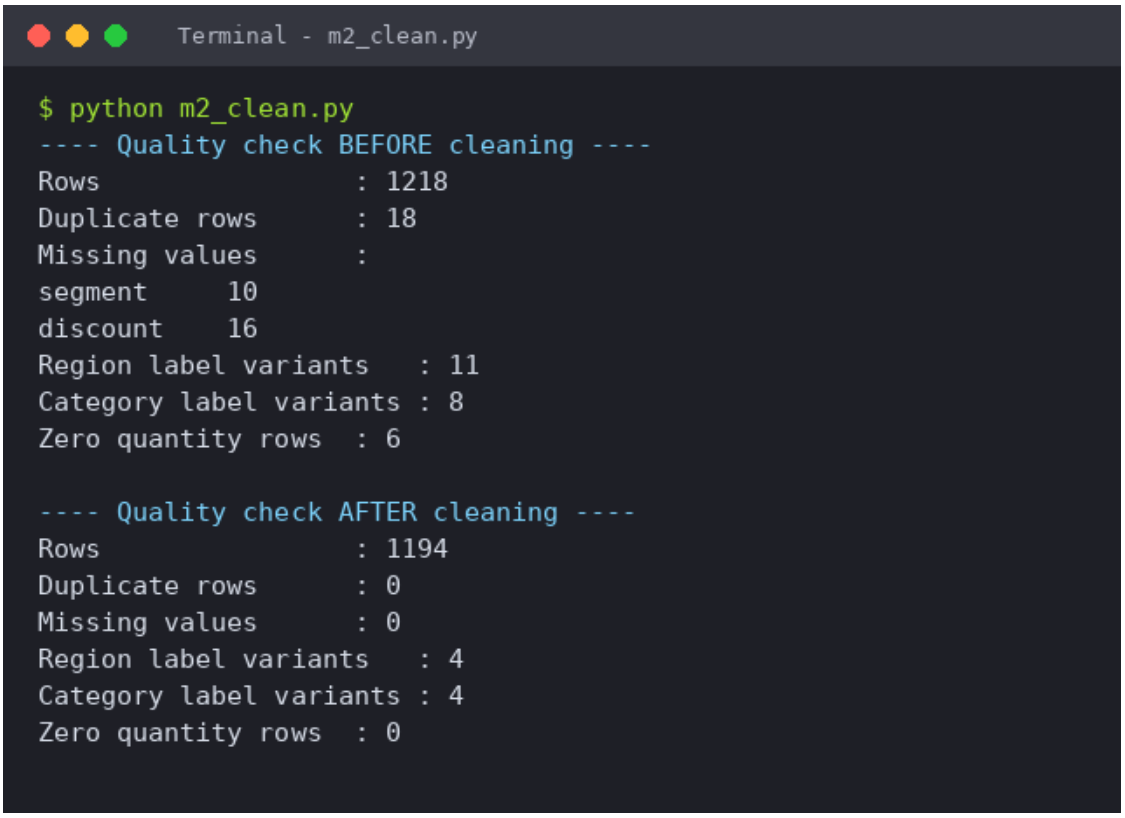


Fig. 4.4 – Data quality check before and after cleaning

The output in Fig. 4.4 shows that 18 duplicate rows and 6 rows with zero quantity were removed, so the number of records fell from 1,218 to 1,194. The 11 spellings of the four regions and the 8 spellings of the four categories were reduced to 4 and 4. No missing values remain, and the cleaned file was saved as clean_sales.csv for the later modules.

4.3 Module 3: Sales KPIs

Services used: Python 3, pandas.

This module reads the clean file and calculates the main indicators of the business: total sales, total profit, profit margin, number of orders, units sold, average order value and number of unique customers. It also prints the descriptive statistics of the numeric columns using the describe function (Code 4.4).

```
import pandas as pd

df = pd.read_csv("data/clean_sales.csv", parse_dates=["order_date"])
total_sales = df["sales"].sum()
total_profit = df["profit"].sum()
orders = df["order_id"].nunique()
units = df["quantity"].sum()

print(f"Total sales (Rs.)   : {total_sales:,.2f}")
print(f"Total profit (Rs.)  : {total_profit:,.2f}")
print(f"Profit margin       : {total_profit / total_sales * 100:.2f} %")
print(f"Number of orders    : {orders}")
print(f"Units sold          : {units}")
print(f"Average order value : {total_sales / orders:,.2f}")
print(f"Unique customers    : {df['customer_id'].nunique()}")
print()
num = ["unit_price", "quantity", "discount", "sales", "profit"]
print(df[num].describe().round(2).to_string())
```

Code 4.4 – Calculation of sales indicators (m3_kpi.py)

```
Terminal - m3_kpi.py

$ python m3_kpi.py
Total sales (Rs.)   : 5,589,069.15
Total profit (Rs.)  : 1,325,630.60
Profit margin       : 23.72 %
Number of orders    : 1194
Units sold          : 3462
Average order value : 4,680.96
Unique customers    : 257

       unit_price  quantity  discount     sales   profit
count     1194.00   1194.00   1194.00   1194.00  1194.00
mean      1786.03      2.90      0.07   4680.96  1110.24
std       1455.84      1.41      0.08   4839.94  1102.73
min        149.00      1.00      0.00    119.20 -2199.60
25%        329.00      2.00      0.00   1023.52   285.78
50%       1799.00      3.00      0.05   2995.00   719.10
75%       2499.00      4.00      0.10   7090.91  1595.58
max       5499.00      5.00      0.30  27495.00  6048.90
```

Fig. 4.5 – Sales indicators and descriptive statistics

The output in Fig. 4.5 shows total sales of Rs. 5,589,069.15 and a total profit of Rs. 1,325,630.60, which is a profit margin of 23.72 %. The average order value is Rs. 4,680.96. The statistics table also shows that the minimum profit is negative, which means that some orders were sold at a loss. The reason for this is studied in Module 6.

4.4 Module 4: Sales Trend Analysis

Services used: Python 3, pandas, Matplotlib.

The aim of this module is to see how sales changed during the year. The sales and profit are grouped by month and by quarter, and the month-over-month growth is calculated with the pct_change function (Code 4.5). A line chart is drawn for sales and profit.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/clean_sales.csv", parse_dates=["order_date"])
order = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"]

monthly = df.groupby("month_name")[["sales", "profit"]].sum()
monthly = monthly.reindex(order).round(2)
monthly["growth_%"] = (monthly["sales"].pct_change() * 100).round(1)
print(monthly.to_string())
print()
quarterly = df.groupby("quarter")[["sales", "profit"]].sum().round(2)
print(quarterly.to_string())
best = monthly["sales"].idxmax(); worst = monthly["sales"].idxmin()
print(f"\nBest month : {best} (Rs. {monthly.loc[best,'sales']:,.2f})")
print(f"Weakest    : {worst} (Rs. {monthly.loc[worst,'sales']:,.2f})")
```

```
fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(order, monthly["sales"] / 1e3, marker="o", color="#2E5C9A", label="Sales")
ax.plot(order, monthly["profit"] / 1e3, marker="s", color="#E07B39", label="Profit")
ax.set_title("Monthly Sales and Profit, 2025"); ax.set_ylabel("Rs. thousand")
ax.grid(alpha=0.3); ax.legend()
plt.tight_layout(); plt.savefig("figs/monthly_trend.png", dpi=150)
```

Code 4.5 – Monthly and quarterly trend analysis (m4_trend.py)

```
Terminal - m4_trend.py

$ python m4_trend.py
                sales     profit  growth_%
month_name
Jan         276773.20   64201.19       NaN
Feb         363152.45   99913.05      31.2
Mar         259475.40   64350.54     -28.5
Apr         460254.45  119506.98      77.4
May         495129.75  111485.47       7.6
Jun         460412.65  105202.48      -7.0
Jul         399129.30   83354.82     -13.3
Aug         428801.35  107827.66       7.4
Sep         455130.65   95880.39       6.1
Oct         614704.95  145275.97      35.1
Nov         603094.10  146310.20      -1.9
Dec         773010.90  182321.85      28.2

              sales     profit
quarter
Q1        899401.05  228464.78
Q2       1415796.85  336194.93
Q3       1283061.30  287062.87
Q4       1990809.95  473908.02

Best month : Dec (Rs. 773,010.90)
Weakest    : Mar (Rs. 259,475.40)
```

Fig. 4.6 – Monthly sales, profit, growth and quarterly totals

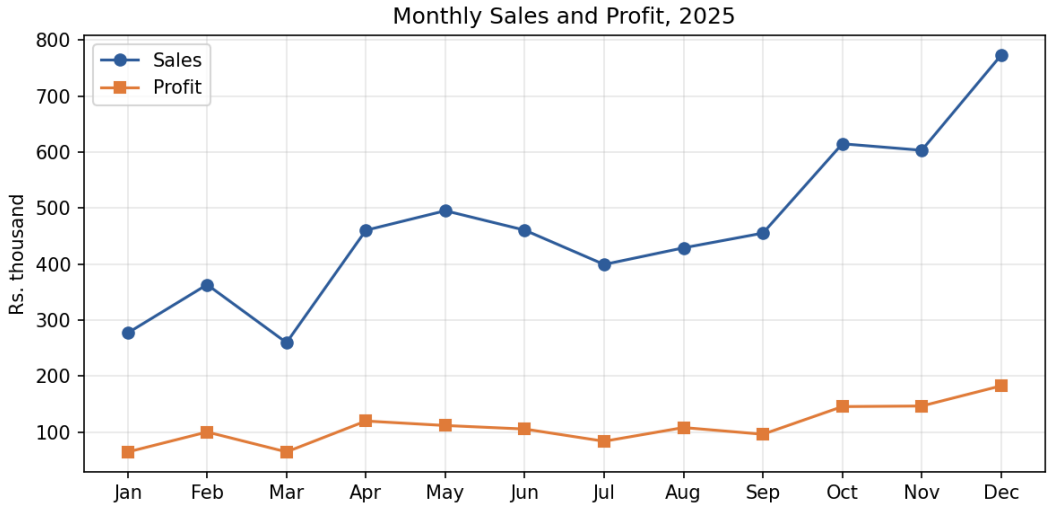


Fig. 4.7 – Monthly sales and profit trend for 2025

Fig. 4.7 shows that sales rise through the year with a sharp increase in the last quarter. Dec is the best month with sales of Rs. 773,010.90 and Mar is the weakest month with Rs. 259,475.40.

4.5 Module 5: Category, Product and Region Analysis

Services used: Python 3, pandas, Matplotlib.

This module finds which categories, products and regions bring the most sales and profit. The profit margin of each category and the percentage share of each region are also calculated (Code 4.6).

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/clean_sales.csv")
cat = df.groupby("category")[["sales", "profit"]].sum()
cat = cat.sort_values("sales", ascending=False)
cat["margin_%"] = (cat["profit"] / cat["sales"] * 100).round(2)
print(cat.round(2).to_string()); print()

prod = df.groupby("product")[["sales", "profit", "quantity"]].sum()
top = prod.sort_values("sales", ascending=False).head(8)
print(top.round(2).to_string()); print()

reg = df.groupby("region")[["sales", "profit"]].sum()
reg = reg.sort_values("sales", ascending=False)
reg["share_%"] = (reg["sales"] / reg["sales"].sum() * 100).round(1)
print(reg.round(2).to_string())

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
cat["sales"].div(1e3).plot.bar(ax=axes[0], color="#2E5C9A", rot=0)
axes[0].set_title("Sales by Category"); axes[0].set_ylabel("Rs. thousand")
top["sales"].div(1e3).sort_values().plot.barh(ax=axes[1], color="#E07B39")
axes[1].set_title("Top 8 Products by Sales"); axes[1].set_xlabel("Rs. thousand")
plt.tight_layout(); plt.savefig("figs/category_product.png", dpi=150)

fig, ax = plt.subplots(figsize=(5, 4))
ax.pie(reg["sales"], labels=reg.index, autopct="%1.1f%%", startangle=90,
       colors=["#2E5C9A", "#E07B39", "#5BA37A", "#9B7FC0"])
ax.set_title("Regional Share of Sales")
plt.tight_layout(); plt.savefig("figs/region_share.png", dpi=150)
```

Code 4.6 – Category, product and region analysis (m5_product.py)

```
Terminal - m5_product.py

$ python m5_product.py
                sales     profit  margin_%
category
Electronics  2014728.30  459577.86     22.81
Furniture    1854214.70  330963.30     17.85
Clothing     1519265.35  465082.39     30.61
Stationery    200860.80   70007.05     34.85

                     sales     profit  quantity
product
Office Chair        793505.70  120098.16       157
Smart Watch         611647.05  113771.55       166
Hooded Jacket       544977.95  163895.71       259
Sports Shoes        530830.35  144512.37       206
Bookshelf           515633.70   98178.24       171
Bluetooth Speaker   510545.70  132696.90       216
Headphones          436981.40  114822.56       237
Study Table         414638.55   71578.35       105

            sales     profit  share_%
region
South   2046392.15  472961.62     36.6
West    1503510.60  359145.57     26.9
East    1193291.15  290609.46     21.4
North    845875.25  202913.95     15.1
```

Fig. 4.8 – Category, top product and region summaries

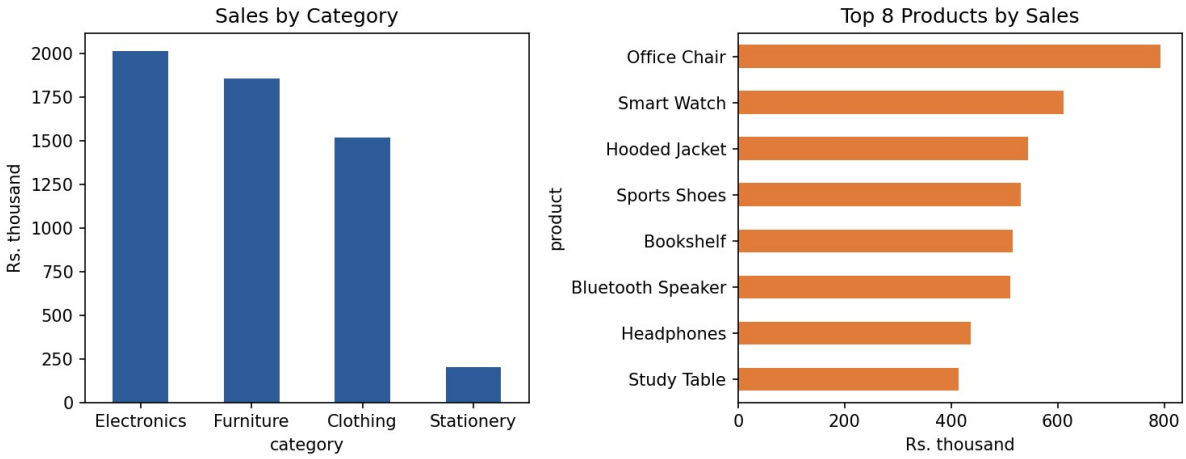


Fig. 4.9 – Sales by category and top eight products by sales

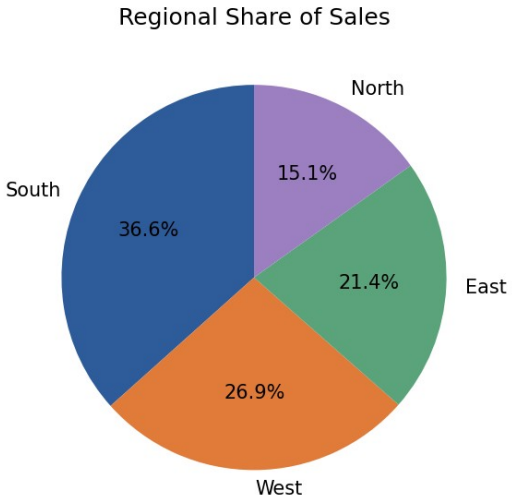


Fig. 4.10 – Regional share of total sales

Fig. 4.9 shows that Electronics is the largest category by sales, but Fig. 4.10 and the tables show that the South region contributes the largest share of the sales. The relationship between region and category is shown in the heat map of Module 7.

4.6 Module 6: Customer, Segment and Discount Analysis

Services used: Python 3, pandas, Matplotlib.

This module looks at the customers. The orders are grouped by customer segment, the discount is divided into four bands with the pandas cut function and the profit margin of every band is calculated, and the five customers with the highest spending are listed (Code 4.7).

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/clean_sales.csv", parse_dates=["order_date"])
seg = df.groupby("segment").agg(orders=("order_id", "count"),
                                sales=("sales", "sum"),
                                profit=("profit", "sum")).round(2)
seg["avg_order"] = (seg["sales"] / seg["orders"]).round(2)
print(seg.to_string()); print()

# discount band versus profit margin
bands = ["0 %", "1-10 %", "11-20 %", "21-30 %"]
df["disc_band"] = pd.cut(df["discount"], [-0.01, 0, 0.10, 0.20, 0.30],
                         labels=bands)
db = df.groupby("disc_band", observed=True)[["sales", "profit"]].sum()
db["margin_%"] = (db["profit"] / db["sales"] * 100).round(2)
print(db.round(2).to_string()); print()

# simple customer value ranking
cust = df.groupby("customer_id").agg(orders=("order_id", "count"),
                                     spent=("sales", "sum"),
                                     last=("order_date", "max"))
print("Top 5 customers by spending")
cust["spent"] = cust["spent"].round(2)
print(cust.sort_values("spent", ascending=False).head(5).to_string())

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
seg["sales"].div(1e3).plot.bar(ax=axes[0], color="#5BA37A", rot=0)
axes[0].set_title("Sales by Customer Segment")
axes[0].set_ylabel("Rs. thousand")
```

```
db["margin_%"].plot.bar(ax=axes[1], color="#9B7FC0", rot=0)
axes[1].set_title("Profit Margin by Discount Band")
axes[1].set_ylabel("Margin (%)")
plt.tight_layout()
plt.savefig("figs/segment_discount.png", dpi=150)
```

Code 4.7 – Segment, discount and customer analysis (m6_customer.py)

```
Terminal - m6_customer.py

$ python m6_customer.py
              orders       sales     profit  avg_order
segment
Consumer         588  2766025.95  645505.62    4704.13
Corporate        383  1735344.30  411098.49    4530.93
Home Office      213  1042682.75  257355.34    4895.22
Unknown           10    45016.15   11671.15    4501.62

                sales     profit  margin_%
disc_band
0 %        2080404.30  621943.88     29.90
1-10 %     2410374.55  565969.71     23.48
11-20 %     915059.00  142998.52     15.63
21-30 %     183231.30   -5281.51     -2.88

Top 5 customers by spending
             orders     spent       last
customer_id
C0133             7  76898.25 2025-11-02
C0166            11  58741.60 2025-12-25
C0021             6  57672.70 2025-12-27
C0120             6  53956.40 2025-06-08
C0247             8  53245.45 2025-11-27
```

Fig. 4.11 – Segment summary, discount-band margins and top customers

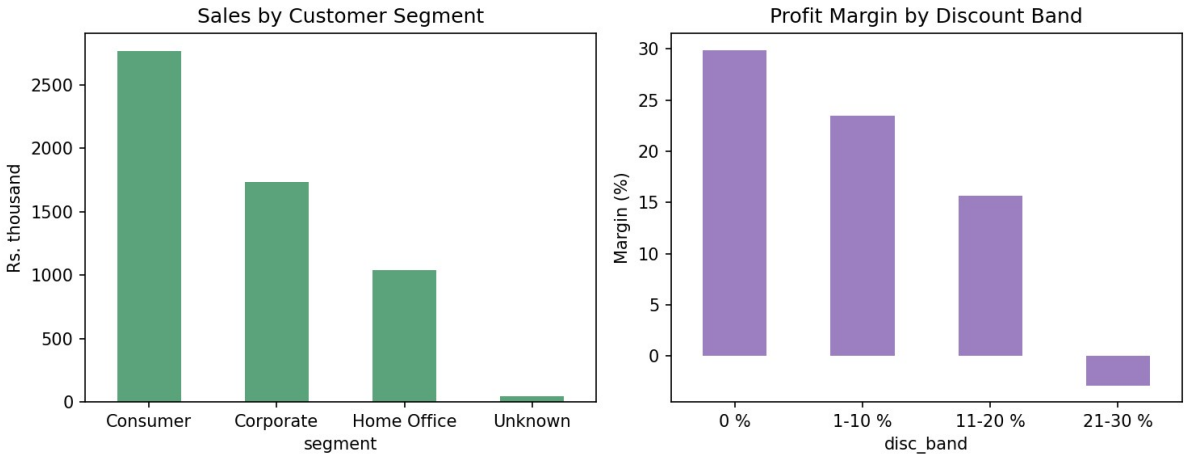


Fig. 4.12 – Sales by customer segment and profit margin by discount band

Fig. 4.12 shows an important result: the profit margin falls steadily as the discount increases, and in the highest band (21–30 %) the margin becomes negative.

4.7 Module 7: Dashboard and Excel Report

Services used: Python 3, pandas, Matplotlib, Seaborn, openpyxl, Microsoft Excel.

The last module prepares the final deliverables. A heat map of sales by region and category is drawn with Seaborn, and a four-panel dashboard is built with Matplotlib so that the manager can see the monthly trend, the category sales, the regional share and the most profitable products on one page. The summaries are also written to an Excel workbook with five sheets – Summary, Monthly, Category, Region and Products (Code 4.8).

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("data/clean_sales.csv", parse_dates=["order_date"])
order = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
         "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
colors = ["#2E5C9A", "#E07B39", "#5BA37A", "#9B7FC0"]

# heat map : region x category
pivot = df.pivot_table(index="region", columns="category",
                       values="sales", aggfunc="sum") / 1e3
fig, ax = plt.subplots(figsize=(7, 3.8))
sns.heatmap(pivot, annot=True, fmt=".0f", cmap="Blues", ax=ax,
            cbar_kws={"label": "Rs. thousand"})
ax.set_title("Sales by Region and Category")
ax.set_xlabel("")
ax.set_ylabel("")
plt.tight_layout()
plt.savefig("figs/heatmap.png", dpi=150)
plt.close()

# four-panel dashboard
fig, axes = plt.subplots(2, 2, figsize=(11, 7))
fig.suptitle("Sales Analytics Dashboard - 2025", fontsize=15, fontweight="bold")
m = df.groupby("month_name")["sales"].sum().reindex(order) / 1e3
axes[0, 0].plot(order, m, marker="o", color=colors[0])
axes[0, 0].set_title("Monthly sales (Rs. thousand)")
c = df.groupby("category")["sales"].sum().sort_values() / 1e3
c.plot.barh(ax=axes[0, 1], color=colors[1])
axes[0, 1].set_title("Sales by category (Rs. thousand)")
r = df.groupby("region")["sales"].sum()
axes[1, 0].pie(r, labels=r.index, autopct="%1.0f%%", startangle=90,
               colors=colors)
axes[1, 0].set_title("Share of sales by region")
p = df.groupby("product")["profit"].sum().sort_values(ascending=False).head(6)
(p / 1e3).sort_values().plot.barh(ax=axes[1, 1], color=colors[2])
axes[1, 1].set_title("Top 6 products by profit (Rs. thousand)")
for a in (axes[0, 0], axes[0, 1], axes[1, 1]):
    a.grid(alpha=0.25)
plt.tight_layout(rect=[0, 0, 1, 0.95])
plt.savefig("figs/dashboard.png", dpi=150)
plt.close()

# Excel report with several sheets
def by(col):
    return df.groupby(col)[["sales", "profit"]].sum().round(2)

kpi = pd.DataFrame({
    "KPI": ["Total sales", "Total profit", "Orders", "Units sold"],
    "Value": [round(df.sales.sum(), 2), round(df.profit.sum(), 2),
              df.order_id.nunique(), int(df.quantity.sum())]})
with pd.ExcelWriter("output/Sales_Report.xlsx", engine="openpyxl") as xw:
    kpi.to_excel(xw, sheet_name="Summary", index=False)
    by("month_name").reindex(order).to_excel(xw, sheet_name="Monthly")
```

```
    by("category").to_excel(xw, sheet_name="Category")
    by("region").to_excel(xw, sheet_name="Region")
    by("product").sort_values("sales", ascending=False).to_excel(
        xw, sheet_name="Products")

print("Charts saved in figs/ and Sales_Report.xlsx written")
x = pd.ExcelFile("output/Sales_Report.xlsx")
print("Sheets in the report :", x.sheet_names)
print(pd.read_excel(x, "Summary").to_string(index=False))
```

Code 4.8 – Heat map, dashboard and Excel report (m7_dashboard.py)

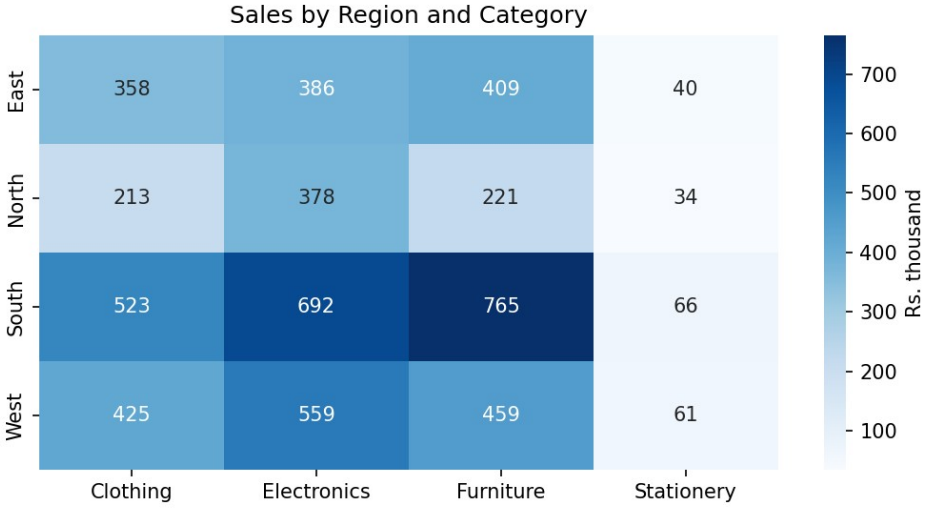


Fig. 4.13 – Heat map of sales by region and category

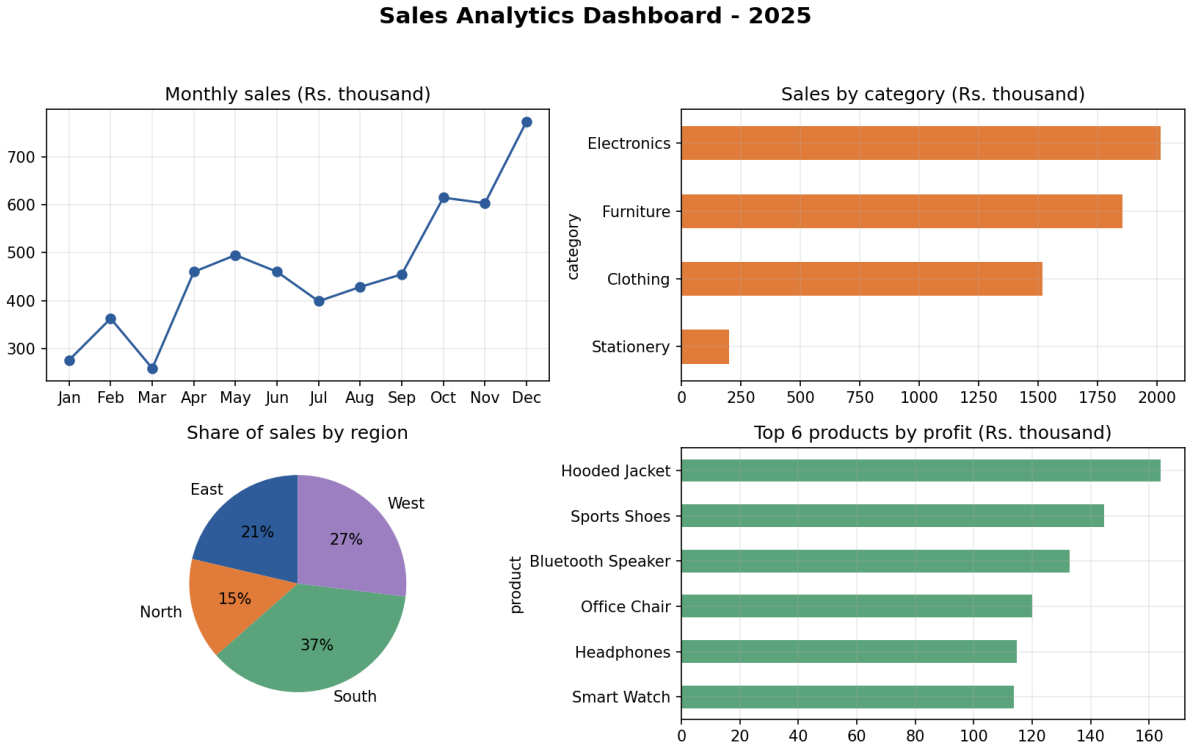


Fig. 4.14 – Sales analytics dashboard for 2025

```
Terminal - m7_dashboard.py

$ python m7_dashboard.py
Charts saved in figs/ and Sales_Report.xlsx written
Sheets in the report : ['Summary', 'Monthly', 'Category', 'Region', 'Products']
         KPI      Value
 Total sales 5589069.15
Total profit 1325630.60
      Orders    1194.00
  Units sold    3462.00
```

Fig. 4.15 – Output of the dashboard and Excel report module

4.8 Testing of the System

The checks of Module 8 were run after all the other modules (Code 4.9). Every test prints PASS or FAIL.

```
import pandas as pd

raw = pd.read_csv("data/raw_sales.csv")
df = pd.read_csv("data/clean_sales.csv", parse_dates=["order_date"])
xl = pd.ExcelFile("output/Sales_Report.xlsx")
total = round(df["sales"].sum(), 2)

def check(no, name, condition):
    print(f"Test {no:02d} {name:<38} {'PASS' if condition else 'FAIL'}")

check(1, "Duplicate rows removed", raw.duplicated().sum() == 18
      and df.duplicated().sum() == 0)
check(2, "Region labels standardised", sorted(df["region"].unique())
      == ["East", "North", "South", "West"])
check(3, "No missing values remain", df.isnull().sum().sum() == 0)
check(4, "Order date is a date column", str(df["order_date"].dtype)
      .startswith("datetime64"))
check(5, "Quantity is always positive", (df["quantity"] > 0).all())
check(6, "Sales never above list price", (df["sales"] <= df["unit_price"]
      * df["quantity"] + 0.01).all())
check(7, "Monthly sales add up to total",
      round(df.groupby("month")["sales"].sum().sum(), 2) == total)
check(8, "Category sales add up to total",
      round(df.groupby("category")["sales"].sum().sum(), 2) == total)
check(9, "Region sales add up to total",
      round(df.groupby("region")["sales"].sum().sum(), 2) == total)
check(10, "Excel report has five sheets", len(xl.sheet_names) == 5)
check(11, "Excel total equals analysed total",
      round(pd.read_excel(xl, "Summary")["Value"][0], 2) == total)
```

Code 4.9 – Automatic consistency checks (m8_tests.py)

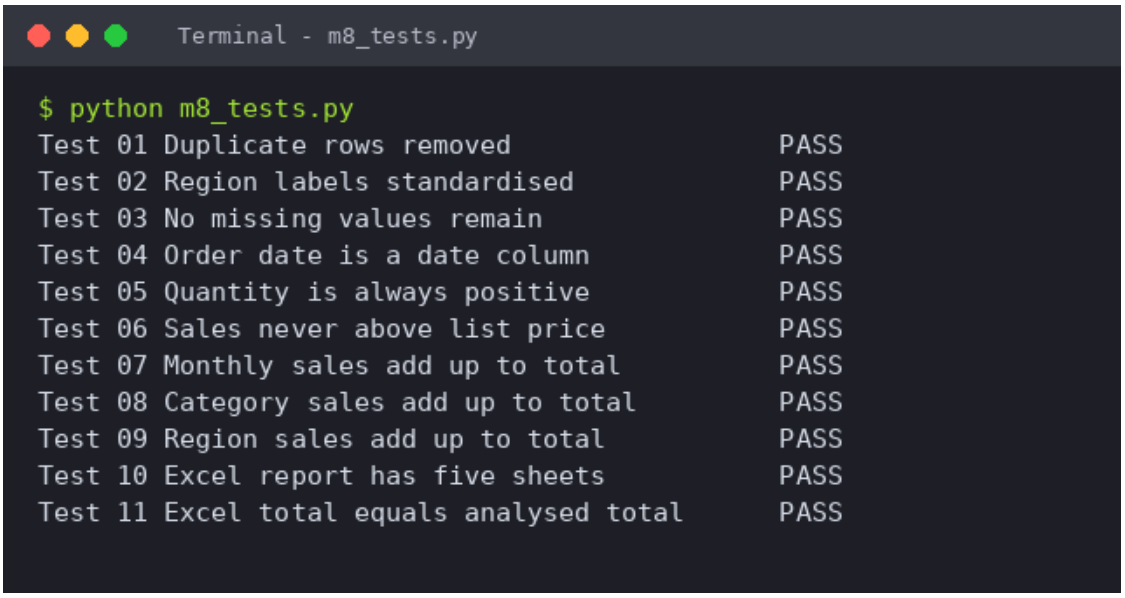


Fig. 4.16 – Result of the automatic checks

Table 4.1 – Test cases and results

No.	Test case	Expected result	Status
1	Remove duplicate rows	18 duplicates in raw file, none in clean file	Passed
2	Standardise region labels	Only East, North, South and West remain	Passed
3	Handle missing values	No missing value in the clean file	Passed
4	Convert order date	order_date is a date column	Passed
5	Remove invalid quantity	Every quantity is greater than 0	Passed
6	Check sales against list price	Sales never exceed unit price × quantity	Passed
7	Monthly totals	Monthly sales add up to the total	Passed
8	Category totals	Category sales add up to the total	Passed
9	Region totals	Region sales add up to the total	Passed
10	Excel sheets	The workbook has five sheets	Passed
11	Excel total	Excel total equals the analysed total	Passed

4.9 Challenges Faced and Solutions

Table 4.2 – Challenges and solutions

Challenge	Solution
Eleven different spellings of four regions and eight of four categories	Spaces were removed with str.strip and the text converted with str.title so that each name has one form
Dates stored as text in day-month-year form	The format “%d-%m-%Y” was given to to_datetime so that the day and month are not confused
Missing discount and segment values	A missing discount was taken as 0 and a missing segment was labelled “Unknown” so that no order was lost

Challenge	Solution
Some orders show a negative profit	These rows were not treated as errors; they were traced to heavy discounts and reported as a finding
Writing sheets to Excel gave an error in the newer pandas version	The sheet name had to be passed as a keyword argument (sheet_name=…) in to_excel
Long code lines did not fit the report page	The scripts were rewritten in shorter lines without changing the results

CHAPTER 5
RESULTS AND DISCUSSION

5.1 Data Cleaning Results

Table 5.1 – Data quality before and after cleaning

Quality issue	Before cleaning	After cleaning
Number of records	1,218	1,194
Duplicate rows	18	0
Missing segment values	10	0 (marked Unknown)
Missing discount values	16	0
Region label variants	11	4
Category label variants	8	4
Rows with zero quantity	6	0

The cleaning removed 24 unusable rows (2.0 % of the raw records) and made every label consistent. All the later results are based on the cleaned data of 1,194 orders.

5.2 Overall Sales Performance

Table 5.2 – Overall sales indicators for 2025

Indicator	Value
Total sales	Rs. 5,589,069.15
Total profit	Rs. 1,325,630.60
Profit margin	23.72 %
Number of orders	1,194
Units sold	3,462
Average order value	Rs. 4,680.96
Unique customers	257

5.3 Monthly and Quarterly Trend

Table 5.3 – Quarter-wise sales and profit

Quarter	Sales (Rs.)	Profit (Rs.)	Share of sales
Q1	899,401.05	228,464.78	16.1 %
Q2	1,415,796.85	336,194.93	25.3 %
Q3	1,283,061.30	287,062.87	23.0 %
Q4	1,990,809.95	473,908.02	35.6 %

Sales in the fourth quarter were Rs. 1,990,809.95, which is 35.6 % of the yearly sales and about 121 % higher than the first quarter. The best month was Dec and the weakest month was Mar. The

pattern suggests a strong festival and year-end season in this sample data set.

5.4 Category-wise Results

Table 5.4 – Category-wise sales, profit and margin

Category	Sales (Rs.)	Profit (Rs.)	Margin
Electronics	2,014,728.30	459,577.86	22.81 %
Furniture	1,854,214.70	330,963.30	17.85 %
Clothing	1,519,265.35	465,082.39	30.61 %
Stationery	200,860.80	70,007.05	34.85 %

Electronics gives the highest sales, but Clothing earns slightly more profit than Electronics on much lower sales because of its higher margin. Furniture has the lowest margin among the large categories (17.85 %). Stationery has the highest margin but contributes very little to the total because its items are low in price.

5.5 Top Products

Table 5.5 – Top eight products by sales

Product	Sales (Rs.)	Profit (Rs.)	Units sold
Office Chair	793,505.70	120,098.16	157
Smart Watch	611,647.05	113,771.55	166
Hooded Jacket	544,977.95	163,895.71	259
Sports Shoes	530,830.35	144,512.37	206
Bookshelf	515,633.70	98,178.24	171
Bluetooth Speaker	510,545.70	132,696.90	216
Headphones	436,981.40	114,822.56	237
Study Table	414,638.55	71,578.35	105

Office Chair is the top product by sales, but Hooded Jacket gives more profit than any other product even though its sales are lower, since its margin is better.

5.6 Regional Results

Table 5.6 – Region-wise sales, profit and share

Region	Sales (Rs.)	Profit (Rs.)	Share of sales
South	2,046,392.15	472,961.62	36.6 %
West	1,503,510.60	359,145.57	26.9 %
East	1,193,291.15	290,609.46	21.4 %
North	845,875.25	202,913.95	15.1 %

The South region contributes 36.6 % of the sales, followed by West (26.9 %). The North region has the lowest share (15.1 %) and is the area with the most room for growth.

5.7 Segment and Discount Analysis

Table 5.7 – Profit margin by discount band

Discount band	Sales (Rs.)	Profit (Rs.)	Margin
0 %	2,080,404.30	621,943.88	29.90 %
1-10 %	2,410,374.55	565,969.71	23.48 %
11-20 %	915,059.00	142,998.52	15.63 %
21-30 %	183,231.30	-5,281.51	-2.88 %

The result is clear: orders sold without discount give a margin of 29.90 %, while orders with a discount of 21–30 % make a net loss (-2.88 %). Among the customer segments, Consumers give the largest sales and Home Office customers have the highest average order value (Rs. 4,895.22). In all, 13 orders made a loss.

5.8 Key Findings and Recommendations

- **Discounts above 20 % should be avoided or restricted,** because they turn profitable products into loss-making orders.
- **Stock and staff should be increased before the October–December period,** which gives more than a third of the yearly sales.
- **Clothing and Electronics should be promoted,** as they give a good balance of sales and profit.
- **Marketing effort in the North region should be increased,** since it has the lowest share of sales.
- **Furniture needs a price or cost review,** because its margin is the lowest of the large categories.

5.9 Skills Developed

Table 5.8 – Skills developed during the internship

Skill area	What was practised
Programming	Writing structured Python programs in modules and running them from the terminal
Data cleaning	Detecting and correcting duplicates, missing values, inconsistent labels and invalid rows
Data analysis	Group-by, pivot tables, descriptive statistics, growth and share calculation
Visualization	Choosing the right chart, labelling charts and building a dashboard
Reporting	Producing an Excel report that can be shared with non-technical readers
Testing	Writing automatic consistency checks to confirm the results
Professional skills	Planning the work in weeks, documenting the project and meeting a deadline

5.10 Comparison with the Manual Spreadsheet Method

Table 5.9 compares the way common tasks are done in a manual spreadsheet with the way they were done in this project.

Table 5.9 – Manual spreadsheet method compared with the Python approach

Task	Manual spreadsheet method	Python approach used
Cleaning labels	Find and replace by hand, different on every file	One rule in code, repeated on every file
Removing duplicates	Visual checking or filter, easy to miss rows	drop_duplicates in one line
Totals by month or region	Pivot table made again for each question	Group-by code that is saved and reusable
Checking the answer	Rarely done	Eleven automatic checks
Charts	Drawn by hand in Excel for every file	Charts are redrawn automatically
Repeating for new data	Repeat all the steps	Run the scripts again

5.11 Internship Experience

The internship at VEI Technologies gave practical exposure to how data science is applied to a business problem. Working step by step from a raw file to a finished report showed that cleaning and checking the data are as important as the analysis itself. Planning the work week by week, writing the code in small modules and testing each result taught good working habits that are useful in any technical job.

5.12 Discussion

The results show that a simple and well-organised analysis can give useful business advice. The most important insight of the project is the effect of discount on profit: the sales total looks healthy in every discount band, but the profit tells a very different story. This could not have been seen by looking at sales alone.

It should be remembered that the project was carried out on a sample data set, so the numerical values are meant to demonstrate the method and not to describe a real company. The same scripts can, however, be applied to a real sales file that has the same columns.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion of Individual Modules

- **Module 1:** the project was organised and the sample data was prepared and inspected.
- **Module 2:** 24 unusable rows were removed and all labels were made consistent, giving a clean file of 1,194 orders.
- **Module 3:** the main indicators were calculated: total sales of Rs. 5,589,069.15, profit of Rs. 1,325,630.60 and a margin of 23.72 %.
- **Module 4:** the sales trend showed a strong last quarter, with Dec as the best month.
- **Module 5:** Electronics led in sales, Clothing in margin and the South region in share.
- **Module 6:** profit margin falls with discount and becomes negative above 20 % discount.
- **Module 7:** a heat map, a dashboard and a five-sheet Excel report were produced, and the checks of Module 8 confirmed that the totals agree.

6.2 Overall Conclusion

The internship project “Sales Data Analysis and Visualization” successfully demonstrated the complete working of a data science project: collecting the data, cleaning it, analysing it, visualizing it and reporting the results. The project achieved all its objectives, and the findings, especially the loss-making effect of heavy discounts, show how data analysis can support real business decisions.

6.3 Future Work

- Connect the scripts to a real company database instead of a CSV file.
- Add sales forecasting for the coming months with time-series or machine learning models.
- Group customers with the RFM method (recency, frequency and monetary value) and with clustering.
- Build an interactive dashboard with Power BI, Tableau or Streamlit so that managers can filter the data themselves.
- Schedule the scripts to run automatically and send the Excel report by e-mail every month.

6.4 Skills Acquired

- Python programming with pandas, NumPy, Matplotlib, Seaborn and openpyxl.
- Data cleaning and validation techniques.
- Exploratory data analysis and business interpretation of results.
- Data visualization and dashboard design.
- Report writing, time management and professional discipline.

6.5 Limitations of the Present Work

- The analysis was carried out on a sample data set of one year, so the numbers cannot be taken

as the figures of a real company.

- A missing discount was taken as zero; this is a simple assumption and may not be true for every row.
- Only descriptive analysis was done; the project does not predict future sales.
- The dashboard is a static image and is not interactive.

6.6 Benefits of the Work

- The scripts can be reused on any new sales file with the same columns.
- The report format helps a manager to understand the business without reading raw data.
- The consistency checks give confidence in the results.
- The project gives the intern a practical base for further study in data science and machine learning.

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APPENDIX
INTERNSHIP COMPLETION CERTIFICATE



VEI TECHNOLOGIES PVT LTD, CHENNAI

(An AICTE, MSME & ISO 9001:2015 Certified Institution)

Corporate Id: 61515598827ee1632720280

VEI TECHNOLOGIES

CERTIFICATE NO: VEI0820261532

CERTIFICATE

OF COMPLETION

This is to certify that Ms. **Sharon Blessy R V** , a student **J.N.N. Institute of Engineering** , has successfully completed an internship in **Data Science** , during which she developed a project titled **"Sales Data Analysis and Visualization"** from 03.08.2026 to 26.08.2026. During this period, she demonstrated commendable dedication, enthusiasm, and professionalism. Her contributions were valuable, and we wish her continued success in her academic and professional pursuits.

DR.B.EZHILAVAN

FOUNDER & CEO

AICTE

MSME

CERTIFIED ISO 9001:2015 COMPANY

Certificate of Completion issued by VEI Technologies Pvt. Ltd., Chennai (Certificate No. VEI0820261532).